1. Clinical Question & Core Hypothesis

Clinical Question

The MGPS idea recognizes that endocrine therapy (ET) response depends on both ER pathway dependence and tumor proliferation, but bulk molecular measures and routine morphology (Ki-67/grade) can be discordant — when and why does this discordance cause failed ET prediction? 2) Tier 1 paths used: the ER-signaling axis (ESR1 and ER-target programs) and the proliferation axis (MKI67/transcriptional proliferation program). I built on these because they form the mechanistic core linking ER dependence to proliferation-driven ET sensitivity or resistance. 3) Tier 2 evidence supporting the claim: ER-target modifiers (GREB1, PGR, TFF1, GATA3) and multiple proliferation genes (CCNB1, AURKA, TOP2A, etc.) provide corroborating transcriptional depth and alternative measures of proliferation that can reveal ESR1–proliferation discordance. 4) Why I chose therapy_response as the endpoint: the stated MGPS objective is to predict endocrine therapy sensitivity/resistance in HR+/HER2- disease, so a therapy-response proxy (ET response / resistance) is the clinically actionable endpoint. 5) What I deliberately EXCLUDED and why: I limited the core hypothesis to exactly one imaging feature and two anchor genes to keep the mechanistic question sharp — I excluded additional genes from the core claim (they appear as secondary_modifiers) and excluded immune imaging features (TIL/TLS) because they are peripheral to ET mechanistic dependence in this MGPS formulation. I also deferred ESR1 mutation status to secondary analyses to avoid conflating de novo ER-pathway activity with acquired resistance mechanisms?

Working Hypothesis

High stromal-to-tumor ratio (TSR) dilutes tumor-cell ER signaling and decouples ESR1-driven transcription from tumor-cell proliferation (MKI67), thereby reducing the predictive value of ER activity for endocrine therapy (ET) response. Consequently, HR+/HER2– breast cancers with low TSR, high ESR1 expression, and low MKI67 will exhibit the strongest ET responses, while high-TSR tumors will show ESR1/MKI67 discordance and diminished ET sensitivity. The strength of ESR1–MKI67 coupling will mediate the TSR–ET outcome relationship, such that TSR interacts with ESR1 and MKI67 status to stratify ET responsiveness beyond ESR1 alone.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	ESR1, MKI67
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ESR1: ESR1 mutation status to secondary analyses to avoid conflating de novo ER-pathway activity with acquired resistance mechanisms.

MKI67: MKI67/transcriptional proliferation program).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	GREB1	Supporting modifier from knowledge graph
Secondary	PGR	Supporting modifier from knowledge graph
Secondary	TFF1	Supporting modifier from knowledge graph
Secondary	GATA3	Supporting modifier from knowledge graph
Secondary	CCNB1	Supporting modifier from knowledge graph
Secondary	AURKA	Supporting modifier from knowledge graph
Secondary	TOP2A	Supporting modifier from knowledge graph
Secondary	BUB1	Supporting modifier from knowledge graph
Secondary	PTTG1	Supporting modifier from knowledge graph
Secondary	KIF2C	Supporting modifier from knowledge graph
Secondary	CENPF	Supporting modifier from knowledge graph
Secondary	NCAPG	Supporting modifier from knowledge graph

Secondary	MAD2L1	Supporting modifier from knowledge graph
Secondary	Ki-67 index (IHC)	Supporting modifier from knowledge graph
Secondary	Histologic grade	Supporting modifier from knowledge graph
Secondary	ESR1 mutation (if available)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective analyses of cohorts with TSR quantified on H&E slides (or validated stromal scores) alongside ESR1 and MKI67 expression and ET outcomes (clinical response, pathologic response, or Ki-67 change).

Prediction 2: Statistical modeling with interaction terms: TSR × ESR1 and TSR × MKI67 to predict ET response, and assessment of ESR1–MKI67 concordance across TSR strata.

Prediction 3: In vitro validation using 3D tumor–stroma co-cultures with variable stromal content to measure ER target gene (e.g., GREB1, PGR) expression and MKI67 in response to estrogen and anti-estrogen therapy.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- GREB1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PGR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.